

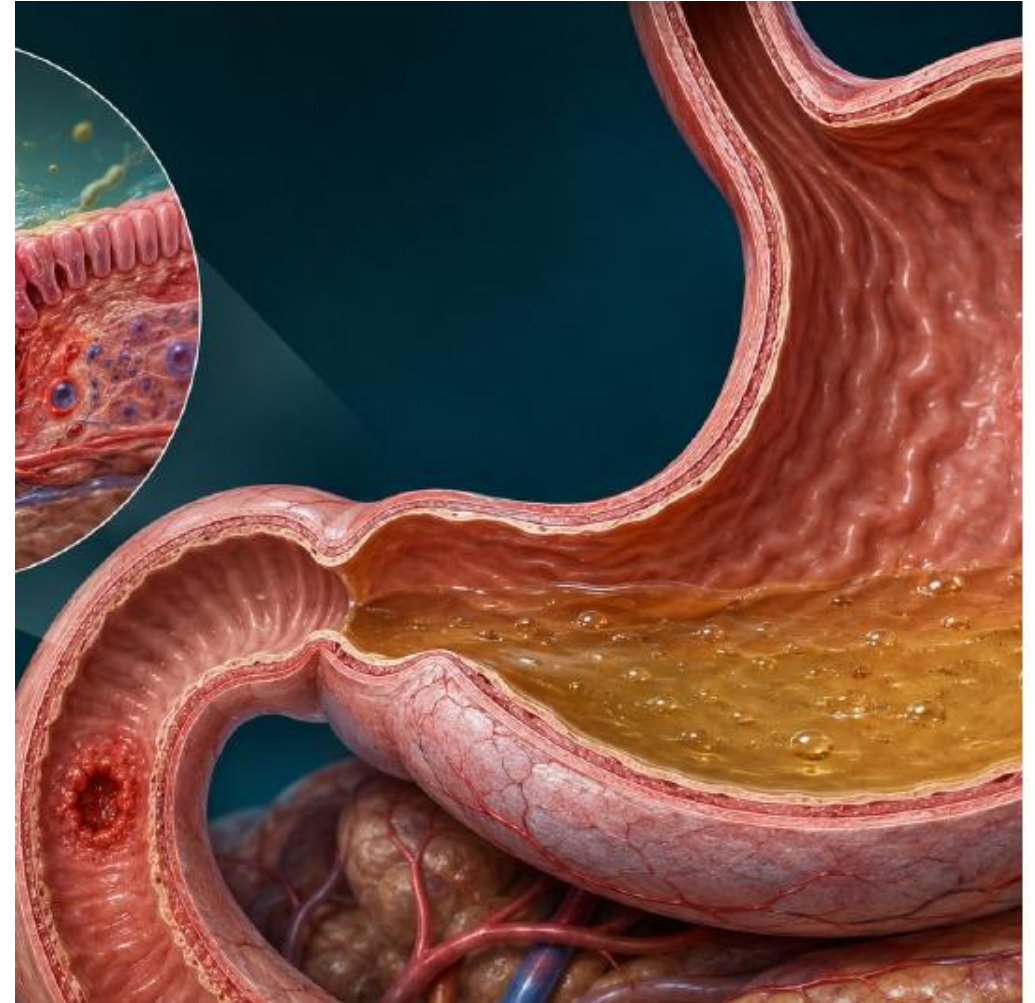
Helicobacter pylori and peptic ulcer

GIT-3rd year
SGL

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Learning objectives

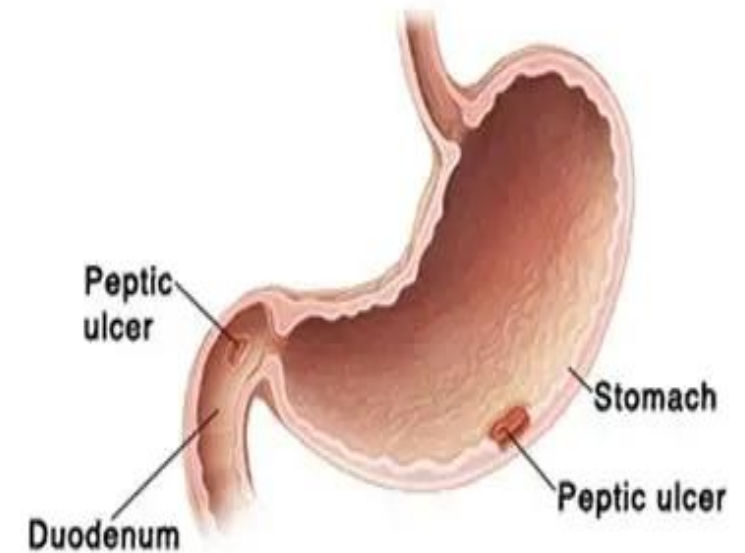
- Describe the morphology, physiology, and growth requirements of *H. pylori*.
- Explain gastric colonization, urease-mediated acid survival, and major virulence factors.
- Relate chronic *H. pylori* gastritis to gastric and duodenal peptic ulcer disease.
- Select and interpret invasive and non-invasive diagnostic tests.
- Outline current eradication therapy, antimicrobial resistance, and test-of-cure principles.

Peptic ulcer definition and main causes

A peptic ulcer is a mucosal defect in the stomach or duodenum produced in an acid-peptic environment. Two major causes dominate clinical practice:

- ❑ *H. pylori* infection
- ❑ Non-steroidal anti-inflammatory drugs (NSAIDs).

H. pylori-associated ulcers are an infectious disease problem: eradication of the organism changes recurrence risk.



Why *H. pylori* matters in medicine

Chronic active gastritis

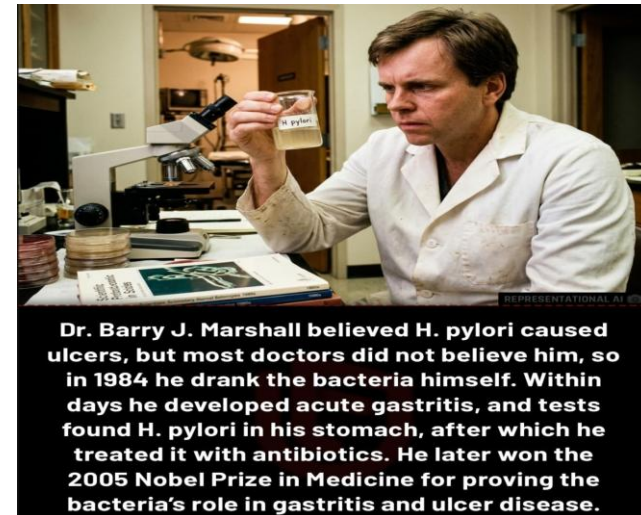
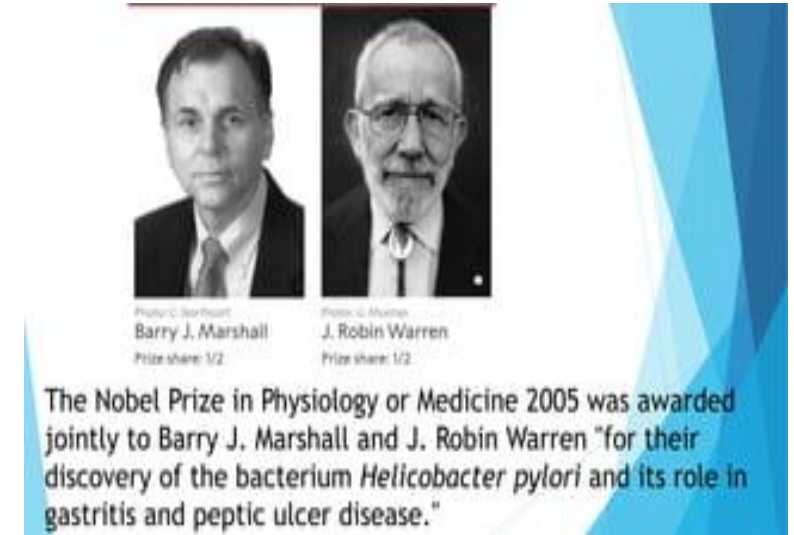
Duodenal and gastric peptic ulcer disease

Gastric adenocarcinoma

Gastric mucosa-associated lymphoid tissue (MALT) lymphoma

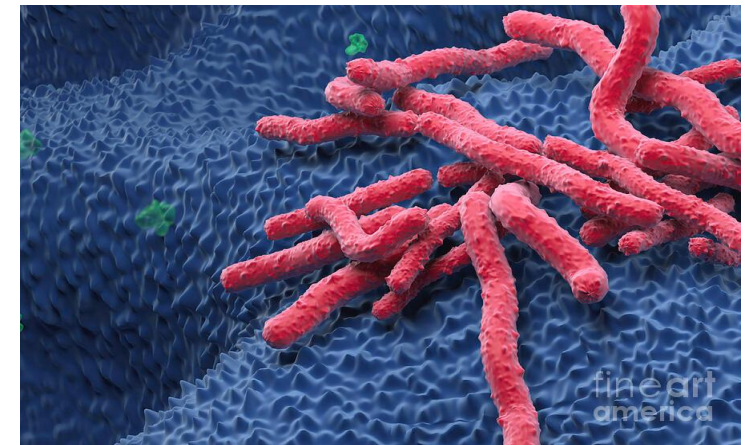
Historical turning point

- Spiral bacteria associated with human gastritis were recognized as a major cause of gastric disease in the **1982**.
- The discovery changed peptic ulcer from an “acid-only” disease to one with a treatable bacterial cause.
- **Marshall** and **Warren** received the 2005 Nobel Prize for the discovery of *H. pylori* and its role in gastritis and peptic ulcer disease.



H. Pylori Morphology

- **Curved**, helical, spiral, Gram-**negative** rod
- Highly motile; multiple **polar flagella** produce corkscrew motility
- **Microaerophilic**
- **Strong urease producer**
- Usually oxidase positive and catalase positive
- Older cultures may show coccoid forms



Growth requirements and culture

Requires enriched media containing blood, serum, charcoal, or similar supplements.

Requires humid microaerophilic incubation, usually 35–37°C.

Colonies may require **several days**; plates may need prolonged incubation before being reported negative.

Culture sensitivity decreases with:

- delayed transport
- prior antibiotics
- suboptimal specimen handling

Reservoir and transmission

H. pylori occurs worldwide.

The **human gastric** mucosa is the major **habitat** of *H. pylori*.

Person-to-person spread is important; **fecal–oral** and **oral–oral** routes.

Once established and untreated, colonization may persist for decades or **for life**.

Most colonized people **do not develop peptic ulcer** or cancer: outcome depends on bacterial, host, and environmental factors.

Major virulence factors

- **Urease** enzyme
acid survival and tissue irritation
- **Flagella**
mucus penetration and colonization
- **Adhesins** (e.g., blood-antigen-binding protein A(BabA)
epithelial attachment
- **Vacuolating cytotoxin A protein** (VacA)
vacuolation and epithelial dysfunction
- **Cytotoxin-associated gene A protein** (CagA)
altered host signaling and inflammation
- **Mucinase**, **phospholipases**, and **inflammatory** mediators
mucosal injury

Pathogenesis and Pathology

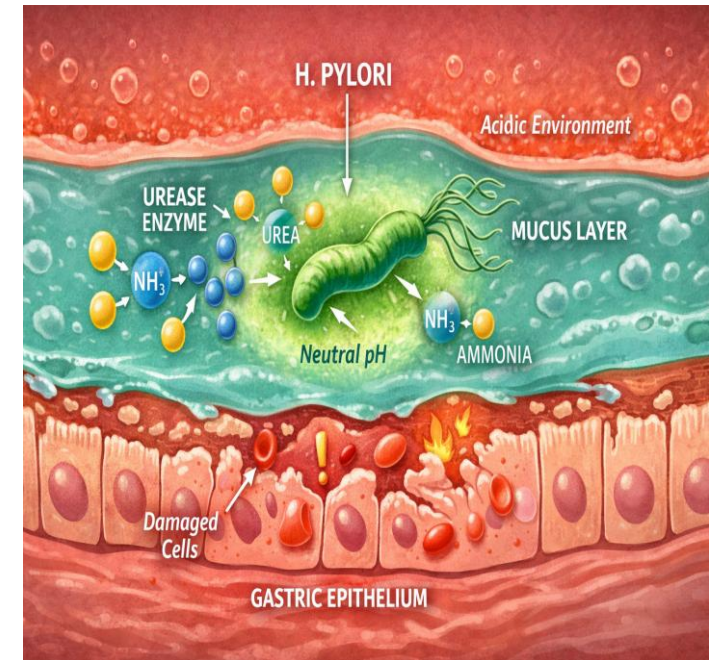
While *H. pylori* grows optimally at a pH of **6.0–7.0**, it would be killed or not grow at the pH within the gastric lumen (pH 1–3).

Several factors contribute to the organism's ability to overcome the acidic environment of the stomach, contributing to colonization, inflammation, changes in gastric acid production, and tissue destruction.

Gastric mucus is relatively **impermeable** to acid and has a strong **buffering** capacity. On the lumen side of the mucus, the pH is low (1.0–3.0); on the epithelial side, the pH is about 5.0–7.0.

After entering the stomach, *H. pylori* utilizes its **urease** activity to neutralize the gastric acid; intracellular urease activity as well as urease located on the bacterial cell surface allow for the breakdown of urea into ammonia and CO₂; NH₃ is converted to ammonium (NH₄⁺) and extruded from the bacterial cell leading to neutralization of the gastric acid.

Flagella-mediated motility then allows the organisms, protected from the gastric acid, to move through the gastric mucus toward the epithelium.

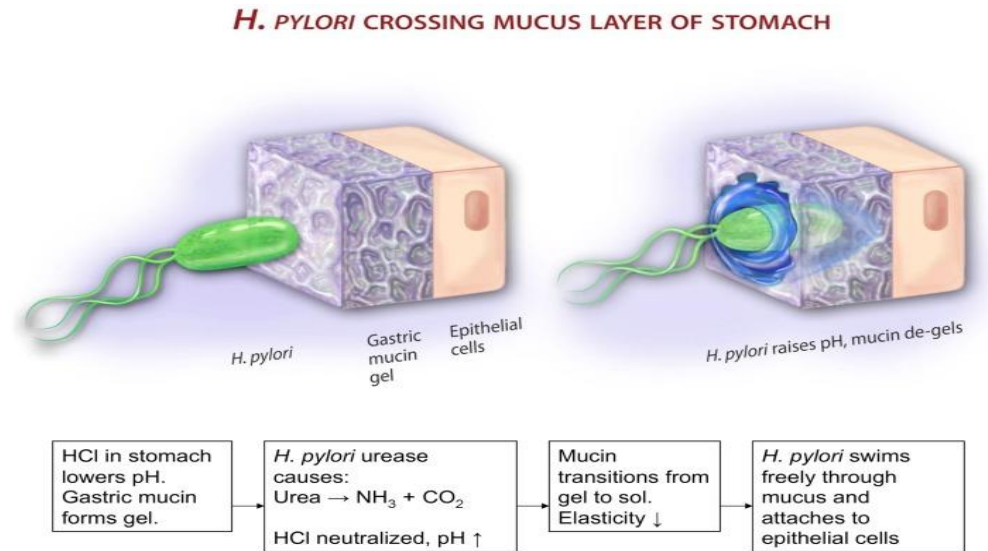


Pathogenesis and Pathology

The actively motile helicobacters can then pass through the gastric mucus and adhere to the gastric epithelial cells by multiple surface **adhesion** proteins.

Localized tissue damage is mediated by **urease by-products**, **mucinase**, **phospholipases**, and the activity of **vacuolating cytotoxin A (VacA)**, a protein that after penetration into epithelial cells damages the cells by producing **vacuoles**.

Another important virulence factor of *H. pylori* is the **cytotoxin-associated gene (*cagA*)**, which interferes with the normal cytoskeletal structure of the epithelial cells



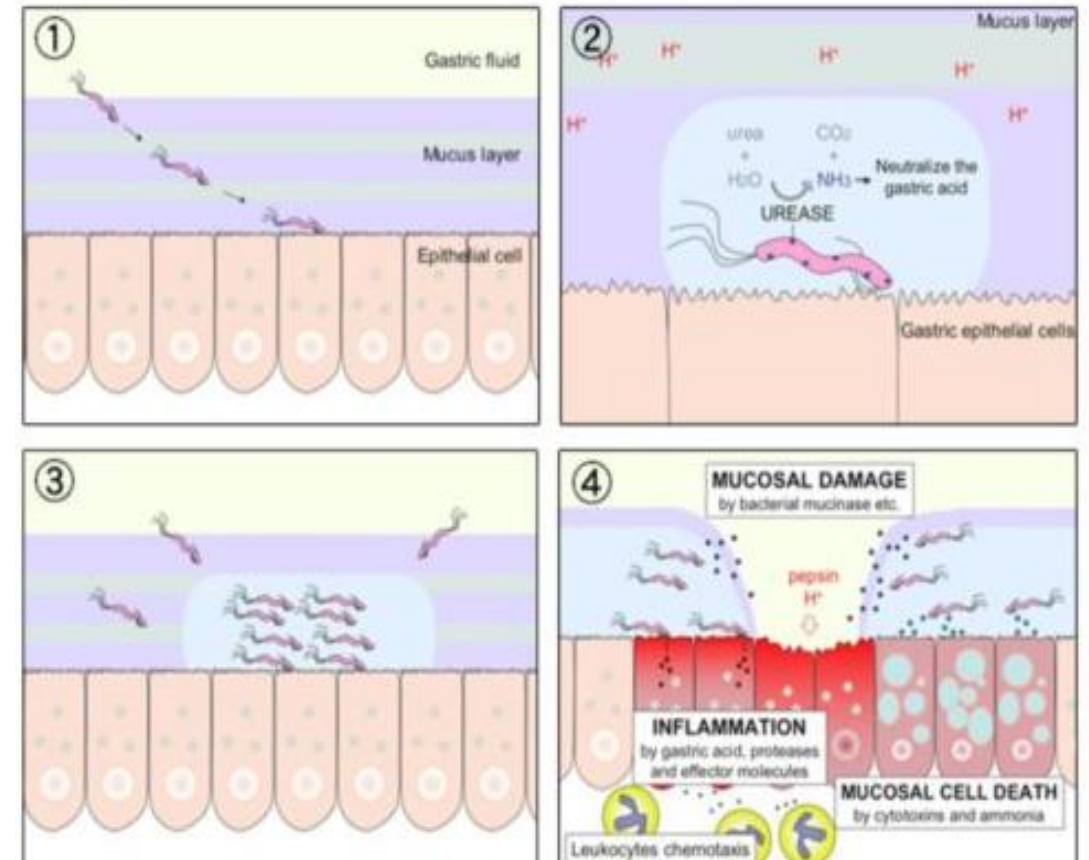
Pathogenesis and Pathology

CagA is secreted by *H. pylori* and then translocated into the gastric epithelial cells via a type IV secretion system; it interferes with the epithelial cells cytoskeletal structure and induces IL-8 production, leading to neutrophil attraction.

VacA affects the balance between cell death and proliferation, and also acts as an activator of IL-8 mediated acute inflammation.

CagA is directly associated with acute gastritis, gastric ulcer development, and gastric carcinoma.

CagA-positive strains are considered more virulent.



Clinical Diseases

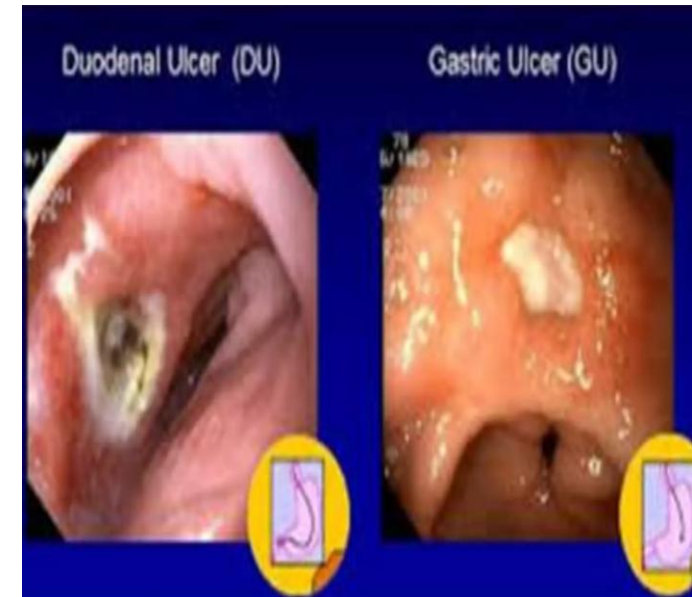
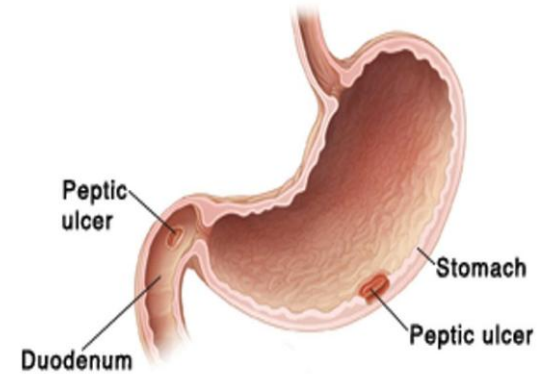
Colonization with *H. pylori* invariably leads to histologic evidence of **gastritis** (i.e., infiltration of neutrophils and mononuclear cells into the gastric mucosa).

The acute phase of gastritis is characterized by:

- feeling of **fullness**
- **Nausea**
- **Vomiting**
- **hypochlorhydria** (decreased acid production in the stomach).

This can evolve into chronic gastritis if acid secretion is suppressed.

Approximately 10% to 15% of patients with chronic gastritis will progress to develop **peptic ulcers**. The ulcers develop at the sites of intense inflammation, commonly involving the junction between the corpus and antrum (**gastric ulcer**) or the proximal duodenum (**duodenal ulcer**).

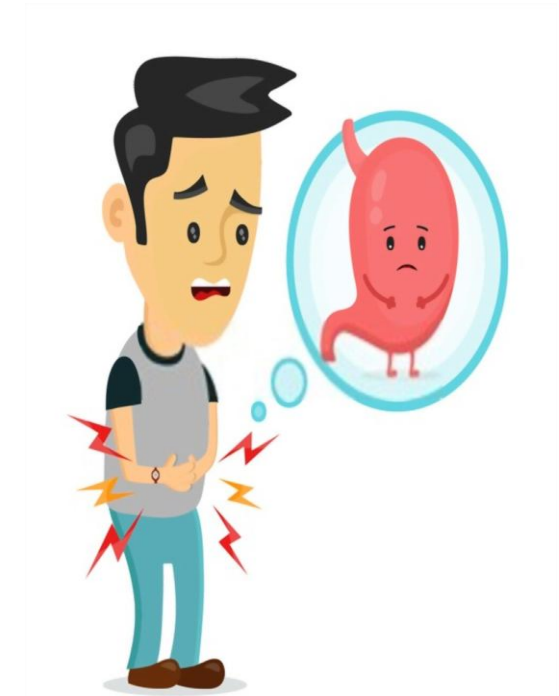


Clinical Diseases

- *H. pylori* is responsible for 85% of the gastric ulcers and 95% of the duodenal ulcers. Recognition of the role of *H. pylori* has dramatically changed the treatment and prognosis of peptic ulcer disease.
- Chronic gastritis eventually leads to replacement of the normal gastric mucosa with fibrosis and proliferation of **intestinal-type** epithelium. This process increases the patient's risk for **gastric cancer** by almost 100-fold. This risk is influenced by the strain of *H. pylori* and the host's response (**cagA-positive strains and high levels of IL-8 production** are associated with a higher risk for cancer).
- Infection with *H. pylori* is also associated with infiltration of lymphoid tissue into the gastric mucosa. In a small number of patients, a monoclonal population of B cells may develop and evolve into a **MALT lymphoma**.

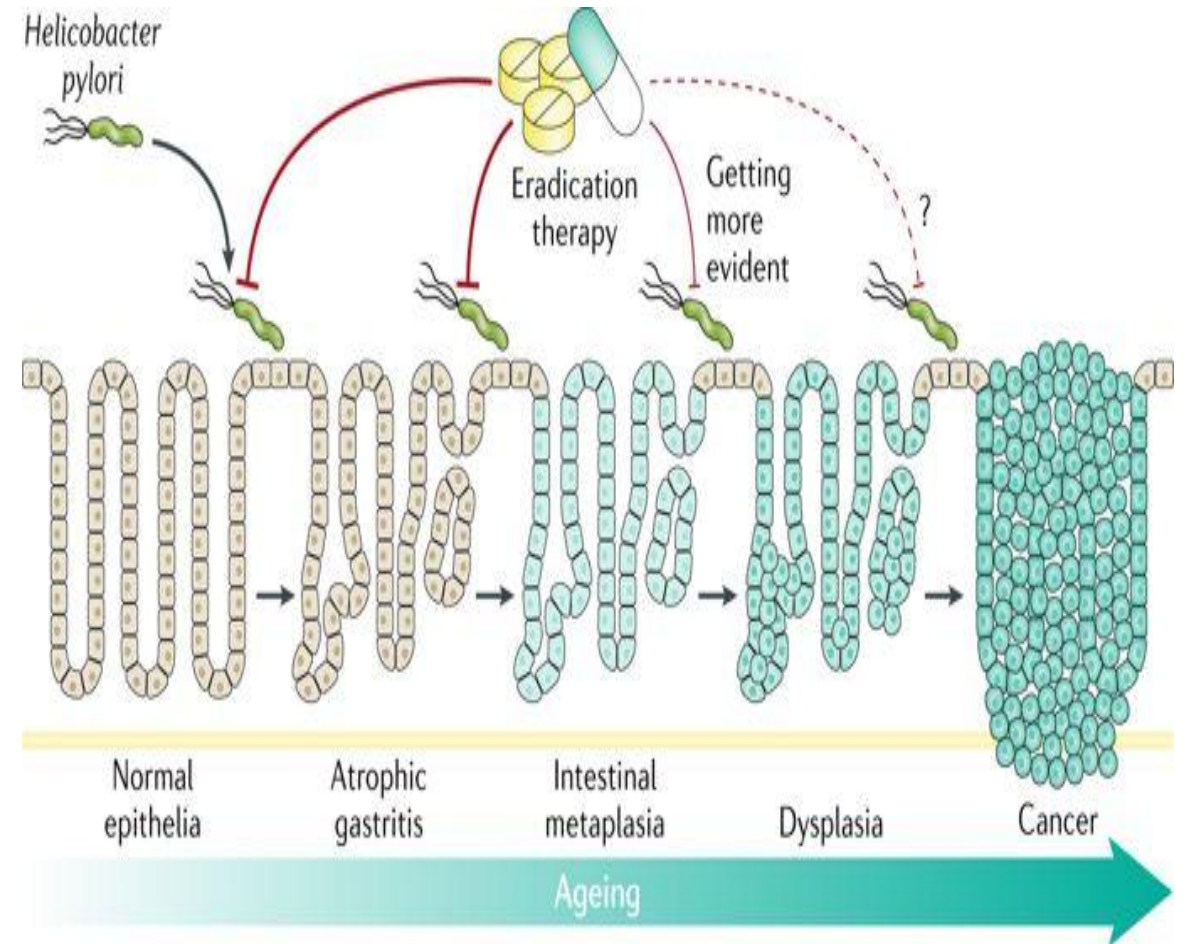
Clinical presentation of peptic ulcer disease

- Epigastric burning
- Dyspepsia, nausea, bloating
- Duodenal ulcer pain may occur when the stomach is **empty** or at night and can improve temporarily with food.
- Gastric ulcer discomfort may **worsen** after meals.
- Symptoms alone cannot reliably determine ulcer location, **endoscopy is definitive.**
- frequently accompanied by bleeding into the gastrointestinal tract.
- No bacteremia or disseminated disease occurs.



Long-term pathway to gastric adenocarcinoma

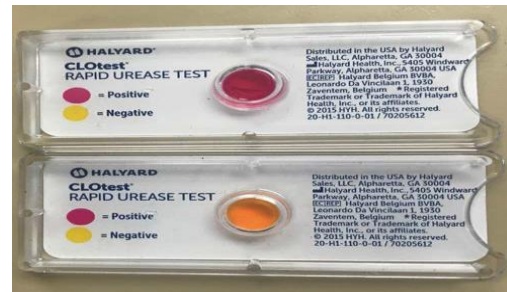
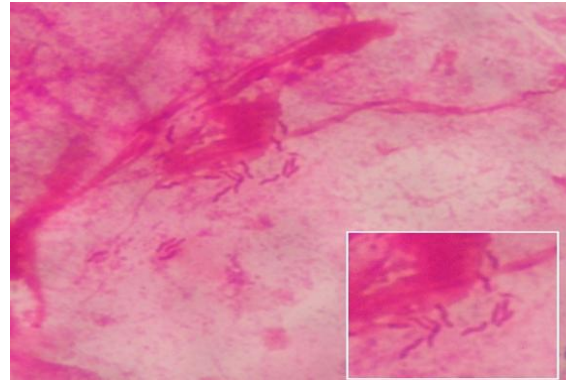
- Persistent *H. pylori* infection
- Chronic active gastritis
- Atrophic gastritis
- Intestinal metaplasia / dysplasia
- Gastric adenocarcinoma



Laboratory diagnosis two pathways

Invasive tests

- Histology of gastric biopsy
- Rapid urease test on biopsy
- Culture with susceptibility testing
- Biopsy-based molecular testing where available

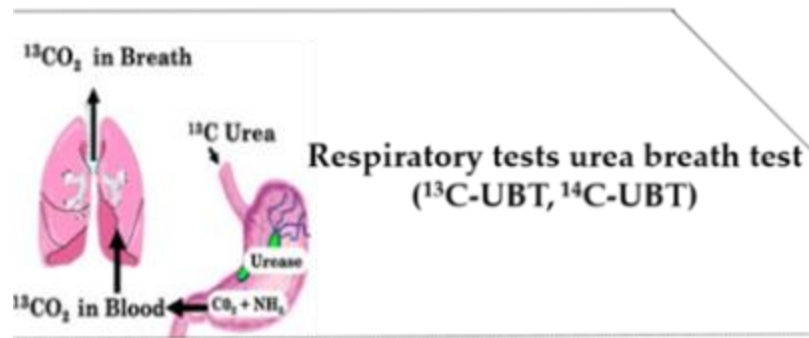


Non-invasive tests

- ^{13}C Urea Breath test
- Stool antigen test
- Serology — demonstrates exposure, not reliable for cure



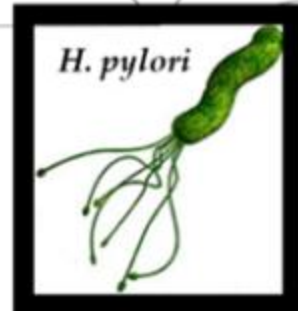
Non-Invasive methods



Stool antigen (SAT)



Current diagnostic tests for *H. pylori*



Invasive methods



Normal



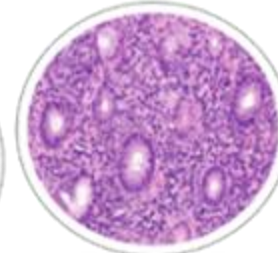
H. pylori gastritis

Endoscopy

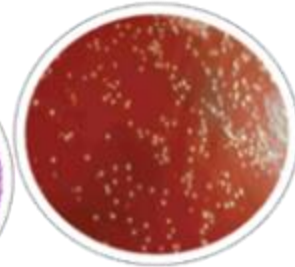
Rapid urease test



Histology



Culture

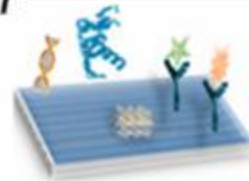
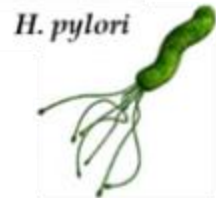


Serology



Biosensors methods

PCR Test



- Optoelectronic nano-biosensors
- Piezoelectric sensors
- Electrochemical biosensors



Detection

False-negative testing common causes

- Recent proton pump inhibitor use
- Recent antibiotics
- Low organism density or patchy colonization
- Inadequate biopsy sampling or improper transport for culture

Laboratory diagnosis

Main limitations

- Biopsy tests require endoscopy
- Culture is slow and technically demanding
- Serology cannot prove active infection or cure

Treatment

The greatest success in curing gastritis or peptic ulcer disease has been accomplished with the combination of:

- **proton pump inhibitor** (e.g., omeprazole)
- **macrolide** (e.g., clarithromycin)
- **β -lactam** (e.g., amoxicillin)
- **metronidazole**

with administration for 7 to 10 days initially.

Treatment failure is most commonly associated with **clarithromycin resistance**. Susceptibility testing should be performed if the patient does not respond to therapy.

Test of cure essential after treatment

- Do not use symptom improvement as proof of eradication.
- Use urea breath test, stool antigen test, or an appropriate biopsy-based test.
- perform test of cure at least 4 weeks after completion of therapy.
- To reduce false negatives, withhold treatment for about 2 weeks before testing when clinically appropriate.

Every treated patient should have eradication confirmed.

MCQ 1

Which bacterial property is most directly responsible for local neutralization of gastric acid around *H. pylori*?

- A. Mucinase activity
- B. Oxidase activity
- C. Urease activity
- D. Catalase activity

MCQ 2

Which test is least appropriate for confirming successful eradication of *H. pylori* after therapy?

- A. ^{13}C Urea Breath test
- B. stool antigen test
- C. biopsy-based test
- D. Serum IgG antibody
- E. None of the above

Case study

A 30-year-old man presents to his physician with complaints of mid-epigastric pain. He describes the pain as moderate, occasionally waking him at night, and improving immediately following meals. A urease breath test was positive. Which of the following correctly describes the causal agent?

- A. Gram-negative curved rod; microaerophilic
- B. Gram-negative rod; aerobic
- C. Gram-negative rod; facultative anaerobe
- D. Gram-positive rod; aerobic
- E. Gram-positive rod; microaerophilic

Take-home messages

- *H. pylori* is a curved, strongly urease-positive, microaerophilic Gram-negative rod adapted to the gastric mucus niche.
- Urease, motility, adhesins, CagA, VacA, and host inflammation drive persistent gastritis and peptic ulcer disease.
- *H. pylori* is also linked to gastric adenocarcinoma and MALT lymphoma.
- Use tests for active infection when diagnosing and confirming eradication.
- Successful treatment requires documented eradication, not symptom relief alone.

References

- Murray PR, Rosenthal KS, Pfaller MA. Medical Microbiology. 10th ed. Elsevier; 2026. Chapter 27: Campylobacter and Helicobacter.
- Riedel S, Morse SA, Mietzner T, Miller S. Jawetz, Melnick & Adelberg's Medical Microbiology. 28th ed. McGraw-Hill; 2019. Chapter 17.
- Chin-Hong P, Joyce EA, Karandikar M, et al. Levinson's Review of Medical Microbiology & Immunology. 18th ed. McGraw Hill; 2024. Chapter 18.
- Tille PM. Bailey & Scott's Diagnostic Microbiology. 16th ed. Elsevier; 2026. Chapter 33: Campylobacter, Arcobacter, and Helicobacter.
- Chey WD, et al. ACG Clinical Guideline: Treatment of Helicobacter pylori Infection. American Journal of Gastroenterology. 2024.
- Malfertheiner P, Megraud F, Rokkas T, et al. Management of Helicobacter pylori infection: the Maastricht VI/Florence consensus report. Gut. 2022;71:1724–1762.